



MODULE - 2 ASS:-2

CYBER SECURITY

I. LIST 3 REAL-WORLD DEVICES IN YOUR HOME OR COLLEGE THAT USE SMPS (SWITCHED MODE POWER SUPPLY) AND 2 DEVICES THAT USE UPS (UNINTERRUPTIBLE POWER SUPPLY), EXPLAINING IN ONE LINE WHY EACH DEVICE NEEDS THAT TYPE OF POWER SUPPLY.



REAL-WORLD DEVICES

- **Devices Using SMPS**

- **Laptop Charger** – Uses SMPS to efficiently convert AC power into the low DC voltage needed by the laptop.
- **LED TV** – Uses SMPS to provide stable and efficient power to its internal circuits.
- **Wi-Fi Router** – Uses SMPS to convert and regulate power efficiently for its electronic components.

- **Devices Using UPS**

- **Desktop Computer** – Uses UPS to provide backup power during power cuts and prevent data loss.
- **CCTV System** – Uses UPS to keep the cameras and recording system working during power interruptions.

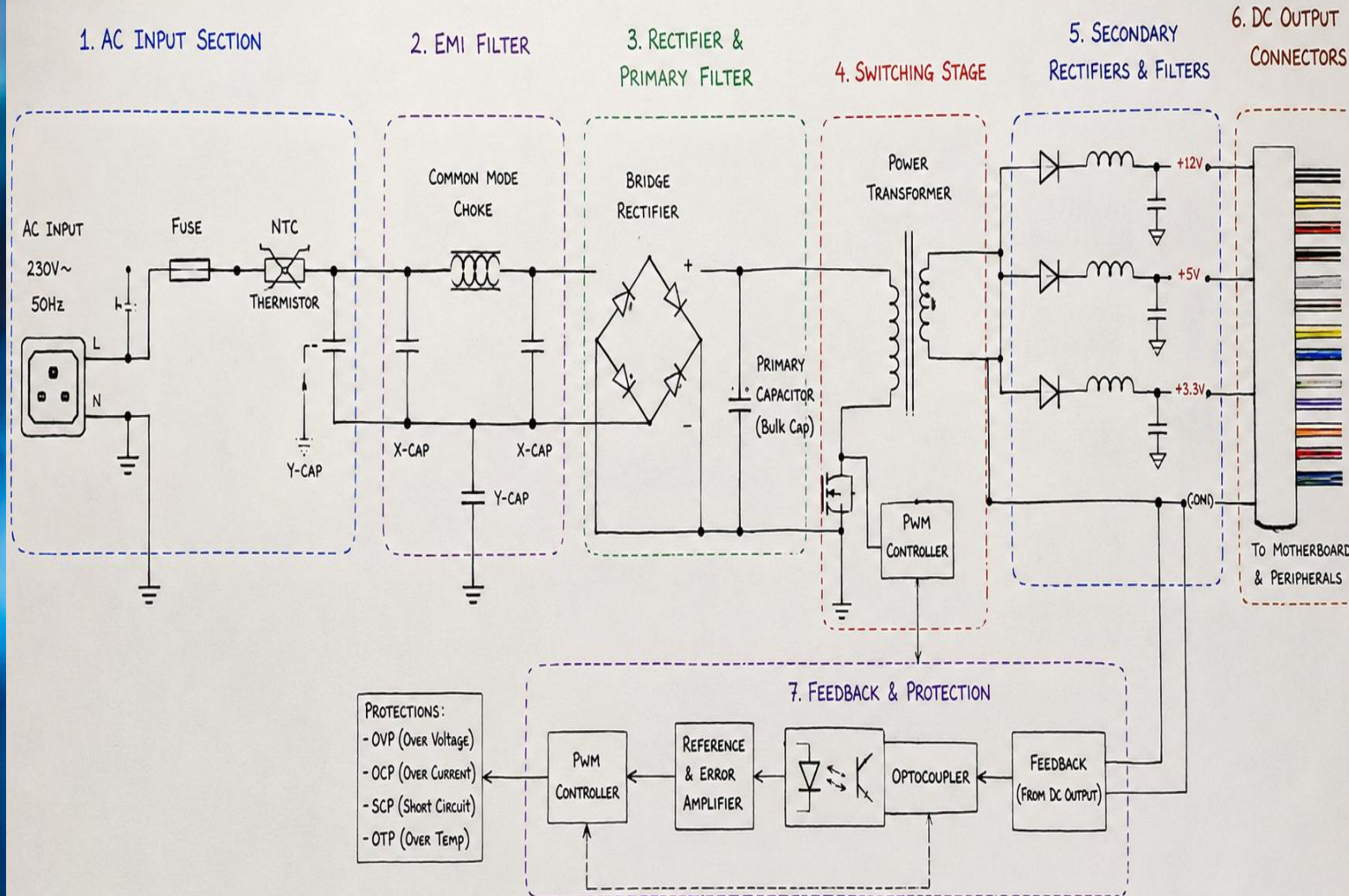
2. DRAW AND LABEL A DIAGRAM SHOWING THE MAIN INTERNAL COMPONENTS OF A TYPICAL SMPS UNIT FOUND IN A DESKTOP PC, INCLUDING AC INPUT, RECTIFIER, TRANSFORMER, AND DC OUTPUT SECTIONS.

HINT: YOU CAN HAND-DRAW AND UPLOAD A PHOTO, OR USE ANY FREE DIAGRAM TOOL LIKE DRAW.IO.



LABEL A DIAGRAM SHOWING THE MAIN INTERNAL COMPONENTS OF A TYPICAL SMPS UNIT FOUND

TYPICAL SMPS (SWITCH MODE POWER SUPPLY) IN A DESKTOP PC

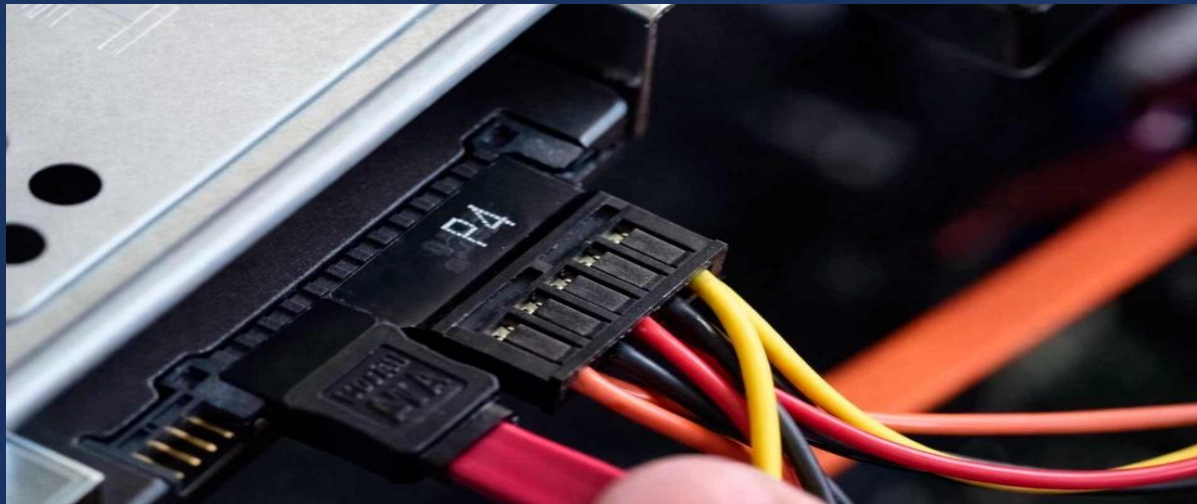
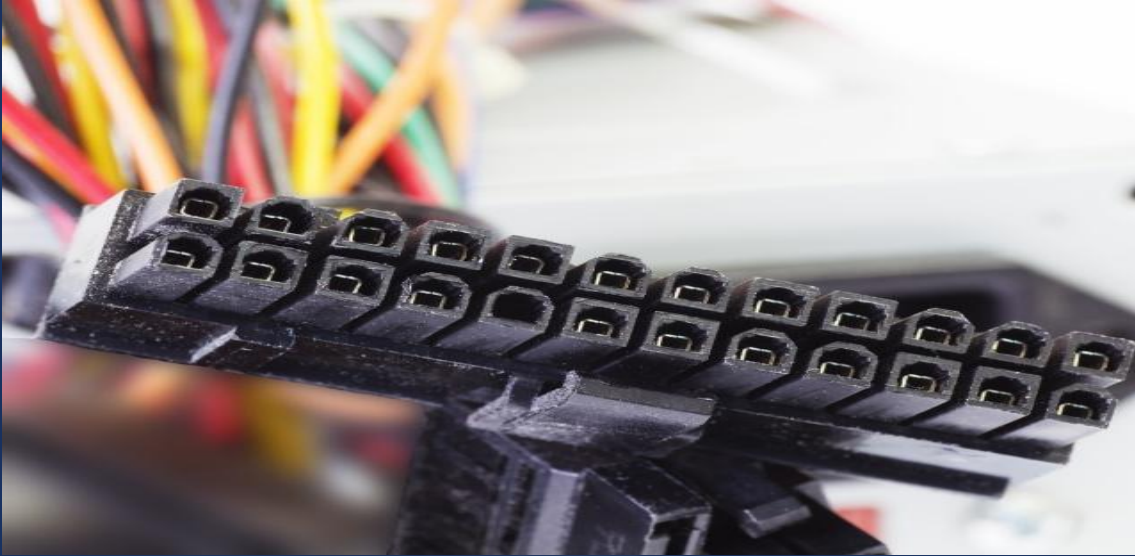


>>DESCRIPTION:

- A typical desktop PC SMPS converts **230 V AC input** into regulated **DC outputs** such as **+12 V, +5 V, and +3.3 V**. First, the AC power passes through a **fuse and EMI filter** for protection and noise reduction. A **bridge rectifier** converts AC into DC, which is then filtered. The high-voltage DC is switched at high frequency by the **switching circuit/PWM controller** and supplied to the **transformer**. The transformer provides isolation and changes the voltage. On the secondary side, **rectifiers and filters** convert the high-frequency output into smooth DC. Finally, the regulated DC is supplied to the **motherboard and other PC components**.

3. TAKE PHOTOS OR FIND CLEAR IMAGES ONLINE OF THE FOLLOWING POWER CONNECTORS: ATX 24-PIN MOTHERBOARD CONNECTOR, SATA POWER CONNECTOR, MOLEX CONNECTOR, AND PCIE 6/8-PIN CONNECTOR. FOR EACH, WRITE ITS NAME, THE DEVICES IT POWERS, AND ONE SAFETY TIP FOR CONNECTING/DISCONNECTING IT.

LABEL A DIAGRAM SHOWING THE MAIN INTERNAL COMPONENTS OF A TYPICAL SMPS UNIT FOUND



>>DESCRIPTION:

Connector

ATX 24-pin Motherboard Connector

SATA Power Connector

Molex Connector (4-pin)

PCIe 6/8-pin Connector

Devices it powers

Powers the **main motherboard**, including its major circuits and slots.

Powers **SATA HDDs, SSDs, optical drives, and some RGB devices**.

Commonly powers **older hard drives, optical drives, case fans, and other peripherals**.

Provides additional power to **dedicated graphics cards (GPUs)**.

Safety tip

Switch off and unplug the PC before connecting or disconnecting, and align the keyed connector correctly—never force it.

Ensure the connector is **properly aligned before inserting** and never pull it out by the wires.

Turn off and unplug the PSU before connecting/disconnecting to avoid short circuits or component damage.

Use **only the PCIe power cable** intended for the graphics card and make sure the locking clip clicks into place.

4. IMAGINE YOU ARE SETTING UP A NEW GAMING PC. LIST THE STEPS YOU WOULD FOLLOW TO CONNECT THE SMPS TO THE MOTHERBOARD, GRAPHICS CARD, AND STORAGE DEVICES, MENTIONING WHICH SPECIFIC CABLES AND CONNECTORS ARE USED FOR EACH.

>> STEPS TO CONNECT SMPS IN A GAMING PC

- **Switch off and unplug the PC** – Make sure the SMPS is disconnected from the mains before starting.
- **Connect the motherboard** – Connect the **24-pin ATX cable** from the SMPS to the motherboard's main **24-pin ATX power socket**.
- **Connect the CPU** – Connect the **4/8-pin EPS (CPU) power cable** from the SMPS to the motherboard's **CPU power connector** near the processor.
- **Connect the graphics card** – Use the **6-pin, 8-pin, or 6+2-pin PCIe power cable** from the SMPS and connect it to the GPU's power socket.
- **Connect SATA storage devices** – Connect a **SATA power cable** from the SMPS to the power port of the **SATA SSD or HDD**.
- **Connect other peripherals** – If required, use **SATA power or Molex connectors** for devices such as case fans, RGB controllers, or older peripherals.
- **Check all connections** – Ensure every connector is fully inserted and its locking clip is secured; do not force any connector.
- **Power on and test** – Connect the SMPS to the mains, switch it on, and start the PC to verify that all components receive power.



THANK YOU

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